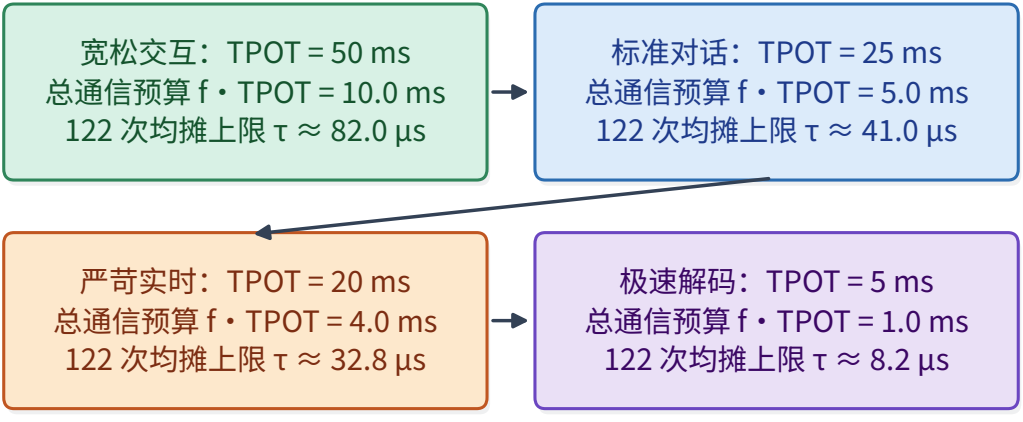
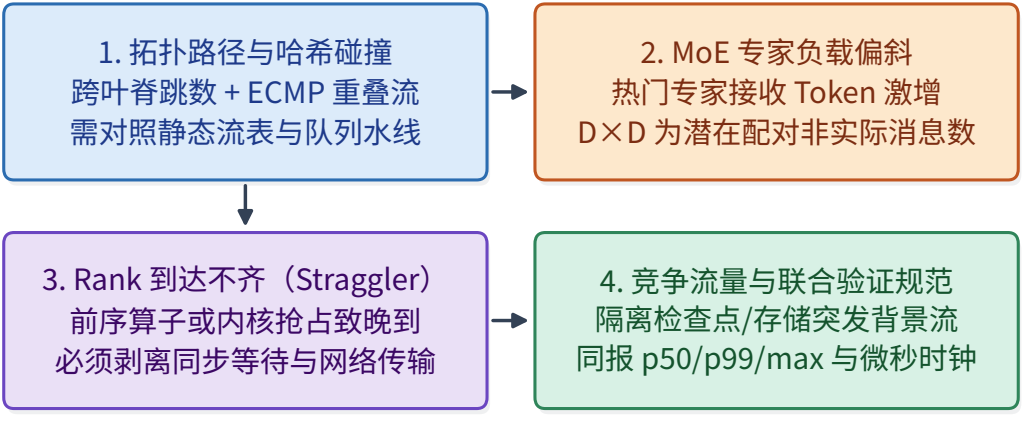
A. 从 TPOT 预算倒推单次集合通信上限（L=61, k=2, f=20%）



B. 集合通信尾延迟四类根因解耦与联合验证矩阵（D11-06）